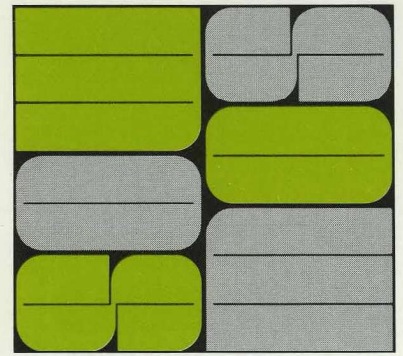


FAIRCHILD SEMICONDUCTOR

MADE IN
FAIRCHILD



SEPTEMBER 1972



FAIRCHILD OPTIMOS

The promise of MOS/LSI is being fulfilled — inexpensive monolithic circuits can replace entire electronic assemblies today. MOS large scale integration processes can provide a subsystem of several hundred to more than a thousand gates at a cost around \$10. Such arrays are now being designed into many more types of digital signal-processing systems than in the past because MOS circuit performance has quadrupled in the past few years.

Nevertheless, MOS/LSI presents risks as well as rewards to the equipment manufacturer. If MOS devices are applicable, all competing equipment manufacturers are virtually compelled to use them. Only the most cost-effective designs, obviously, will be successful. In every instance, since MOS/LSI is a subsystem rather than a component technology, the difference between success and failure will be the ability of the MOS manufacturer to support the cost and design goals of the system planner.

No single approach to MOS/LSI design covers all needs. Striking the right balance among standard MOS products, custom MOS development and bipolar support functions, for example, demands meticulous evaluation of MOS products, design alternatives, and development costs and timing.

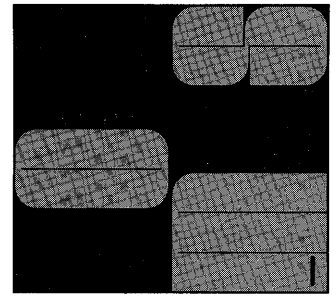
This guidebook will examine those planning factors as they relate to MOS/LSI in general and to OPTIMOS in particular. OPTIMOS, a Fairchild acronym for optimum MOS in terms of capability and application is a method of exercising all major MOS cost and design options. It encompasses all Fairchild resources in standard and custom MOS development and production.

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The OPTIMOS process is graphically represented in the form of a flow chart at the back of this manual.

Where MOS Makes Sense Today



The equipment designer's task of choosing between MOS and bipolar circuits used to be relatively simple. MOS/LSI reigned supreme in low power applications while bipolar SSI/MSI (small-to-medium scale integration) was uncontested wherever high circuit speed was essential. As a result of advances in both MOS and bipolar processing, however, the performance attributes attainable through each have moved closer together. In fact, many systems being designed today can tolerate both MOS speeds and bipolar power requirements as well.

So it almost always makes sense to consider MOS/LSI totally or in part, where there is digital signal processing to be performed. Even where medium-to-high circuit speeds are required, the innovative implementation of MOS/LSI with bipolar SSI/MSI may provide a more economical solution than conventional bipolar designs. Now more than ever, the system planner must carefully analyze design alternatives before settling on any single semiconductor process or combination of processes.

MOS/LSI Lowers Digital Signal Processing Costs

In digital processing applications that do not require critically high circuit speeds or drive capability, MOS/LSI more than bipolar SSI/MSI can potentially lower the overall system development cost. Not only is the MOS transistor smaller than the bipolar transistor, but it is self-isolating as well. This combination of smaller transistor area and virtual elimination of isolation area is a major contributor to the cost-savings realizable through MOS because functional density per unit area on a given chip is far greater than that attainable in bipolar MSI/LSI. Moreover, MOS/LSI more readily lends itself to dynamic circuit techniques, and therefore requires fewer components per function.

MOS/LSI should thus be considered for any digital circuit design, unless the speed performance of bipolar logic is essential. Medium speed systems are sometimes produced at unnecessarily high costs with bipolar logic assemblies because the designer did not realize that in large scale arrays, silicon gate MOS/LSI can equal DTL speed.

The overall system design should be studied to determine which portions must work at bipolar speed and drive levels and which can tolerate MOS speed (up to 3 MHz for random logic and up to 8 MHz for structured logic clock rates such as shift registers). Computer peripheral interface logic and storage functions typically operate at or below MOS speed.

There is also the possibility of employing parallel processing to achieve high throughput with relatively low clock rates. This approach can be highly effective because of extremely low cost per gate of MOS/LSI.

The Buck Stops Here

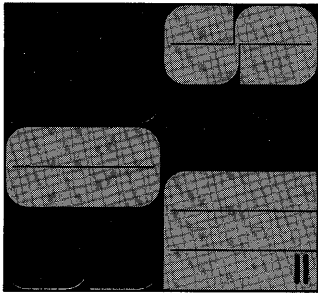
If you are considering your first MOS/LSI system, be prepared to make some hard decisions. To get the best results from the technology may involve some pioneering and some risk.

The conservative approach is not always practical or safe. A small logic system of 300 to 500 gates can be converted to one large scale integrated circuit at a cost of about \$10 per circuit in large production quantities. Beyond that, it starts to get sticky because MOS/LSI makes its own rules about large systems and how they should be sliced.

A system partitioned along lines which have grown to be industry standards will no doubt be more economical than the same system assembled with custom packaged logic — assuming MOS is suitable for the system.

While a system that wholly or partially uses standard parts may provide economics over one designed with custom parts, there always exists the possibility that a competitive manufacturer using a totally custom approach may achieve a product of equal or superior performance at a reduced cost.

Furthermore, MOS technology is still in a state of flux. The rules are not fixed. They change from application to application and with each introduction of a new standard circuit, a new set of custom design cells, or an improvement in wafer fabrication processes.



MOS Economics: An Overview

At one time, MOS circuits were employed in avionics and portable military equipment primarily to achieve small size and low power consumption as well as improved reliability through minimum use of failure-prone component interconnections. The early MOS/LSI circuits were expensive because of poor manufacturing yields.

Now lower manufacturing cost, due to improved yields, is the overriding factor in the phenomenal growth of MOS applications in commercial, industrial, automotive and consumer electronics equipment. User costs have dropped to considerably less than 1¢ per gate in many standard MOS products. In large production quantities, custom MOS circuits can approach standard MOS costs.

Before the prospective MOS user decides how to structure and partition his system — or which process and vendor to use — he should first understand MOS economics. The cost of using MOS typically ranges from purchasing standard products such as a 1024-bit random access memory (RAM) for under \$10 for 1000 pieces to \$100 for a custom design (including amortized design cost) for the same quantity. The user must decide where in this economic range he will derive the maximum benefit from MOS implementation.

Production Volume Governs Custom Cost

The cost of an MOS custom circuit development is governed primarily by the volume of circuits purchased. In other words, the larger the production run, the smaller the development cost per circuit. Accompanying this cost/volume relationship is the proverbial learning curve. As the production volume increases, the manufacturer becomes more familiar with the circuit. Accordingly, fewer processing

errors and higher yields can be expected to further drive down the unit cost.

The potential economics of using custom MOS/LSI versus standard bipolar SSI/MSI can be readily demonstrated through a direct comparison. Consider, for example, a subsystem designed with 300 TTL gates. In assorted MSI and SSI functions, this represents about 30 dual in-line packages. While actual field figures show that the *indirect* cost per bipolar package averages about \$2.00, let's make a conservative comparison and assume this cost to be only \$1.00. This indirect per package cost which covers component insertion, PC board, testing and documentation thus totals \$30 for the bipolar implemented system. To be even more conservative, let's ignore the unit cost of the bipolar devices and assume the total subsystem cost to be \$30.

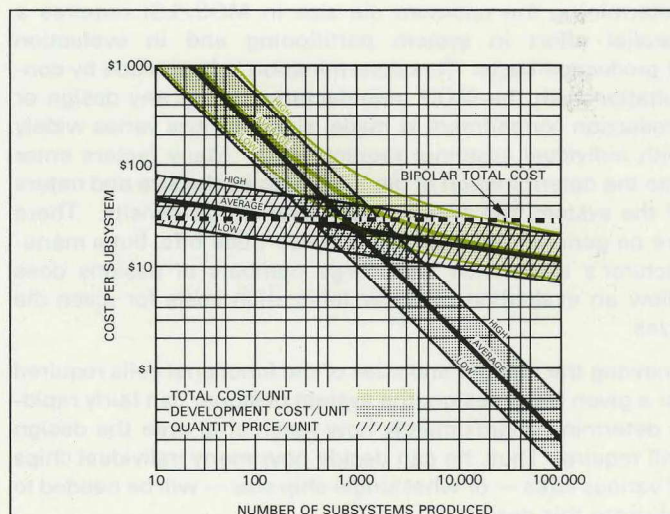
Now if the same subsystem were designed with a single custom MOS circuit, the one-time development charge would be about \$20,000 and the unit cost would be about \$15 in quantities of 10,000. As shown in *Figure 1*, MOS implementation becomes a more economical approach at a production level that is greater than 2100 subsystems. At the 15,000 subsystem level, a two times advantage is held by MOS. Beneath 2100 subsystems, bipolar MSI/SSI is more economical.

The graph in *Figure 1* also depicts a cost spread on the MOS implementation representing higher and lower complexity levels than the 300 gate example. The one-time

development charges could range from \$10,000 to \$50,000. This will depend on the design effort required for logic circuit and layout implementation and test pattern generation. The unit cost could vary from \$10 to \$30 in production quantities of 10,000 depending on the size and complexity of the chip.

The higher cost for smaller quantities of custom MOS subsystems stems chiefly from the need of the MOS manufacturer to amortize custom development and manufacturing start-up costs through either front-end charges to the customer or by a contractual commitment for a large volume of units. In the past, MOS manufacturers were frequently willing to underwrite much of the start-up cost as a means of winning customers. Besides, some of the start-up costs could not be distinguished from the MOS manufacturer's own investments necessary to initiate MOS development and processing.

Since these costs are now well established and the direct result of individual development projects, MOS manufacturers are generally reluctant to undertake a custom development order without a firm commitment regarding payment of costs or a production volume sufficient to amortize the costs. For these reasons, the system manufacturer is advised to make sure before making a decision to go custom that the right custom approach is started — indeed, that a custom approach is more economical for the volume of subsystems required.



The COMPARATIVE COSTS OF A 300-GATE SUBSYSTEM are depicted for both custom MOS/LSI and standard bipolar SSI/MSI approaches. Assuming the total cost of a bipolar implemented subsystem to be \$30, the custom MOS/LSI implementation which is assumed to have a one-time development charge of \$20,000 becomes more economical as the production volume exceeds 2100 subsystems. In an increasing number of cases, the subsystem functions can be performed with standard MOS/LSI circuits, thereby eliminating custom development cost. For example, Fairchild's Digital Voltmeter Logic (3814) is a standard MOS/LSI device that provides a complex measurement function — and yet costs less than the bipolar equivalent at all production levels.

Figure 1. Comparative Costs of a 300-Gate Subsystem

Optimum Die Size Growing Larger

Determining the optimum die size in MOS/LSI requires a parallel effort in system partitioning and in evaluation of production costs. This determination is best made by consultation with the MOS manufacturer before any design or production commitment is made, since die size varies widely with individual customer requirements. Many factors enter into the determination of die size, such as the size and nature of the system, the process used, and circuit density. There are no general rules governing these trade offs, but a manufacturer's experience with large numbers of designs does allow an evaluation of wafer fabrication costs for given die sizes.

Knowing the number and sizes of the functional cells required for a given MOS design, the system designer can fairly rapidly determine approximately how much chip area the design will require. Thus, he can decide how many individual chips of various sizes — or what single chip size — will be needed to fabricate this design.

Consider now the equation for determining die cost:

$$\text{Die Cost} = \frac{\text{Wafer Fabrication Cost}}{(\text{Gross Dice/Wafer}) (\text{Probe Yield})}$$

The *gross dice/wafer* represents the number of potentially good dice, and is dependent on wafer area and die size. The relationship of die size to the number of potentially good dice garnered from a 2-inch diameter wafer is shown below:

Die Size (Mils)	Gross Dice/ 2" Wafer	Die Size (Mils)	Gross Dice/ 2" Wafer
120 x 120	166	200 x 200	47
140 x 140	115	220 x 220	36
160 x 160	84	240 x 240	28
180 x 180	62		

The *probe yield* in the die cost equation is also related to die size as it is the product of two parameters:

- the percentage of semiconductor material that is processed within specifications and independent of die size (typically ranges from 60% to 80%); and
- the defect-limited functional yield that is governed by both the die size and the active area density on the die; this parameter represents the probability that a failure-causing defect will *not* occur on any given die.

Thus as die size increases, the probability that a failure-causing defect will occur on any given die also increases. Accordingly, the probe yield decreases, thereby causing the die cost to increase.

While the die cost equation is helpful in placing the range of die sizes in perspective, it only approximates the cost per function, which, in the final analysis, is the major determinant of optimum die size. *Figure 2a* graphically relates die size to cost per unit of functional area. This analysis concludes that the most functions per dollar would be derived from a die size between 140 x 140 mils and 160 x 160 mils. Of course the curves in *Figure 2a* will change with time; *Figure 2b* projects this change through 1975.

If a subsystem function is too large to fit on an optimum size chip, it may not necessarily be wise to partition the function to operate on two or more smaller chips. Such partitioning

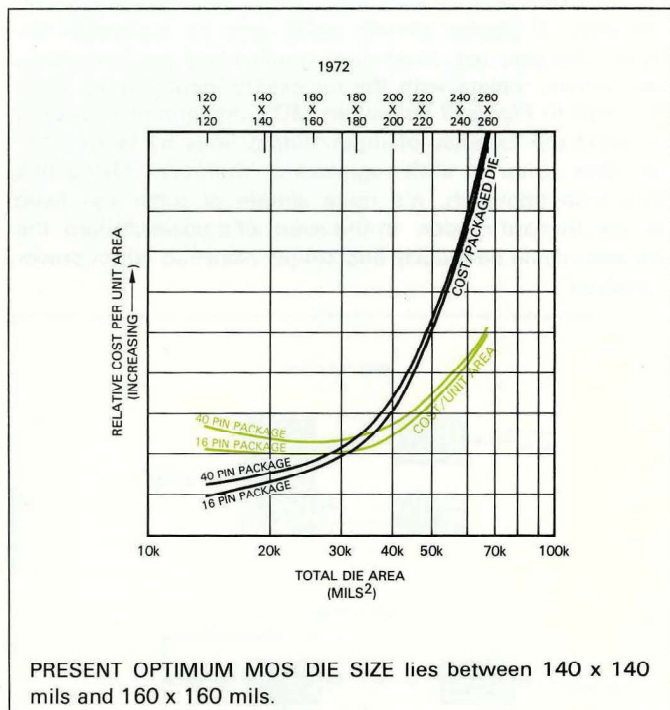


Figure 2a. Present Optimum MOS Die Size

can sometimes complicate the interconnection and logic design or cause performance degradation. Moreover, if the subsystem is to be in production for several years, it is

probably wise to choose a die size that is larger than the current optimum range. The optimum die size has been increasing year by year. By 1975, the optimum die size will likely be 240 x 240 mils. The increasing size of an optimum die coupled with increases in MOS circuit density may entirely eliminate the need to partition a subsystem, indeed an entire system, into two or more smaller dice. Accordingly, the MOS cost per function may drop by a factor of 7 by 1975.

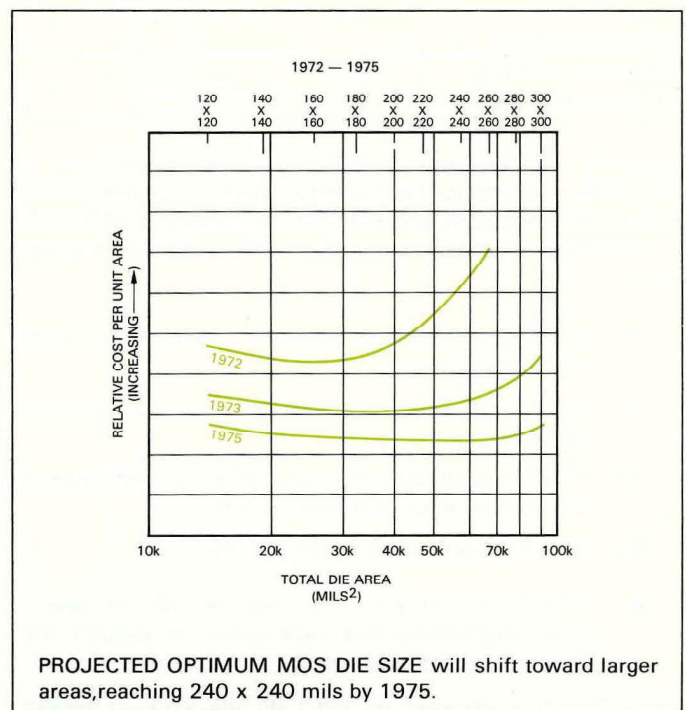


Figure 2b. Projected Optimum MOS Die Size

3257

DOT MATRIX

CHARACTER GENERATOR 64 CHARACTERS 5×7 BITS

FAIRCHILD SILICON GATE MOS INTEGRATED CIRCUITS

GENERAL DESCRIPTION — The 3257 is a Character Generator designed to display 64 characters in a 5 x 7 font. An on chip column select counter sequences through the five columns of each character. The seven output buffers will each drive one TTL/DTL load directly at a 1 MHz input address rate making the 3257 an ideal device for vertical scan CRT displays. The chip enable allows wired-OR capability if more than 64 characters are required.

- **PROGRAMMABLE WITH A CUSTOM CHARACTER FONT**
- **500 ns TYPICAL ACCESS TIME**
- **STANDARD PRODUCT ASCII ENCODED**
- **DIRECT INTERFACING WITH TTL/DTL (FIGURE 8)**
- **WIRED - OR CAPABILITY**

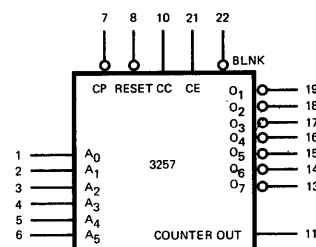
ABSOLUTE MAXIMUM RATINGS — (Above which useful life may be impaired)

Storage Temperature T_S	-65°C to +150°C
Operating Temperature T_A	0°C to +70°C
Voltage on any Pin Relative to V_{SS}	-20 V to +0.3 V

APPLICATIONS:

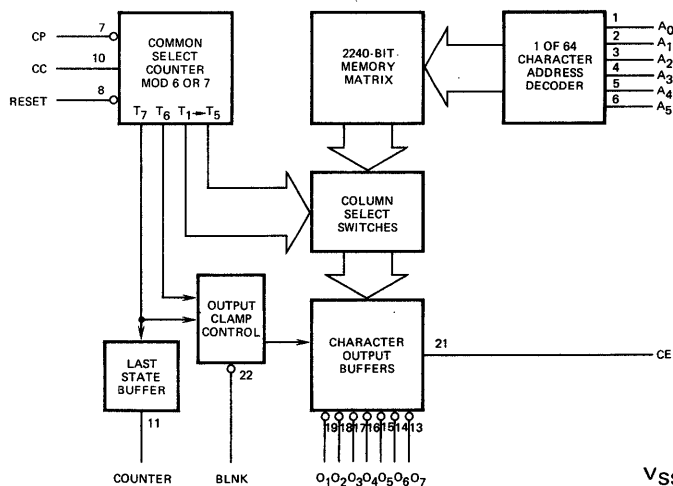
CRT Displays
Billboard Displays
LED Matrix Displays

LOGIC SYMBOL



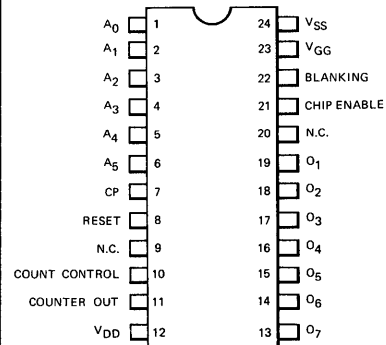
V_{SS} = Pin 24
 V_{GG} = Pin 23
 V_{DD} = Pin 12

BLOCK DIAGRAM



V_{SS} = Pin 24
 V_{GG} = Pin 23
 V_{DD} = Pin 12

CONNECTION DIAGRAM (TOP VIEW)



FUNCTIONAL DESCRIPTION — A Reset pulse ($\sim$ GND) is required to set the counter to the last state. A 6-bit binary word presented to the character address inputs is decoded to select 1 of 64 characters in the memory. Information, representing the first column of the character is available the next clock time after Reset returns HIGH ($\sim$ V_{SS}). The remaining four columns are sequentially selected by the next four states of the counter. The last state of the counter clamps the outputs HIGH ($\sim$ V_{SS}) to provide 1 or 2 space blanking between characters (Count Control $\sim$ V_{SS} $\Rightarrow$ MOD 7, Count Control $\sim$ GND $\Rightarrow$ MOD 6). When the last state (6th or 7th) of the counter is reached, the Counter Output goes HIGH ($\sim$ V_{SS}). When Chip Enable goes HIGH ($\sim$ V_{SS}), the chip is activated while a LOW ($\sim$ GND) at this lead floats the outputs to allow common output bussing. A LOW ($\sim$ GND) on the Blanking input pulls the outputs HIGH ($\sim$ V_{SS}), providing blanking independent of the counter state or the character address.

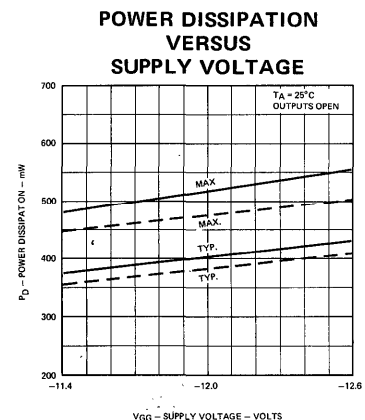
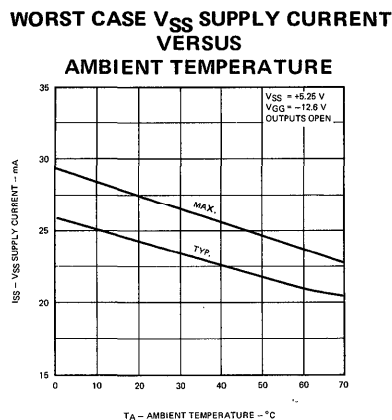
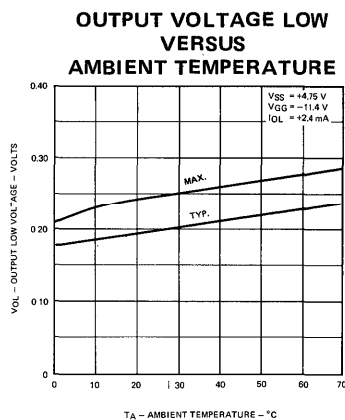
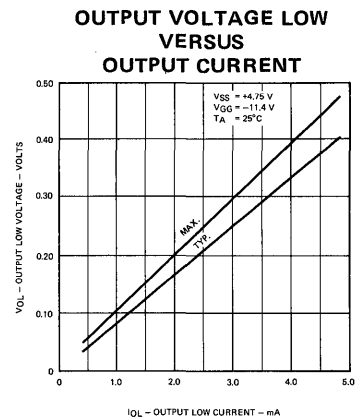
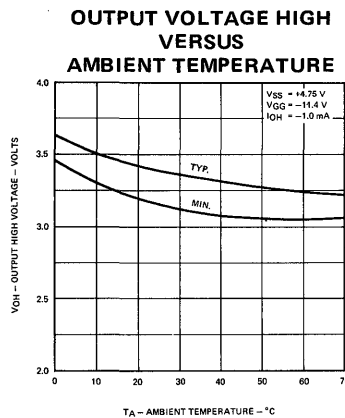
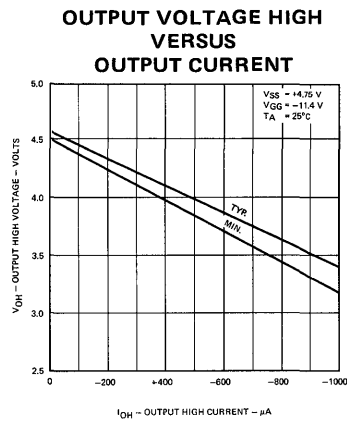
DC CHARACTERISTICS:

V_{SS} = +5 V $\pm$ 5%, V_{GG} = -12 V $\pm$ 5%, V_{DD} = 0 V, T_A = 0°C to +70°C (Standard operating conditions unless otherwise specified)

SYMBOL	CHARACTERISTIC	MIN.	TYP.	MAX.	UNITS	CONDITIONS
V _{IH}	Input Voltage HIGH	V _{SS} - 1		V _{SS}	V	Note 1
V _{IL}	Input Voltage LOW	V _{GG}	0	0.8	V	Note 1
V _{OH}	Output Voltage HIGH	V _{SS} - 0.5 2.4	3.0	V _{SS} V _{SS}	V V	I _{OH} = -10 μ A I _{OH} = -0.5 mA
V _{OL}	Output Voltage LOW	0	0.3	0.4	V	I _{OL} = 1.6 mA
I _{LI}	Input Leakage Current			-1.0	μ A	V _{SS} = 0 V, V _{IN} = -18 V, Note 1
I _{LO}	Output Leakage Current			1.0	μ A	V _{SS} = 0 V, V _{OUT} = -6 V, Note 2
I _{SS}	V _{SS} Current		20	40	mA	V _{SS} = 5.25 V, V _{GG} = -12.6 V Outputs Open
P _D	Power Dissipation		360	715	mW	

NOTES:

1. Inputs include Character Address, Count Control, Clock and Reset.
2. Chip Enable = LOW.
3. I_{SS} = -I_{GG} (V_{GG} Supply Current)

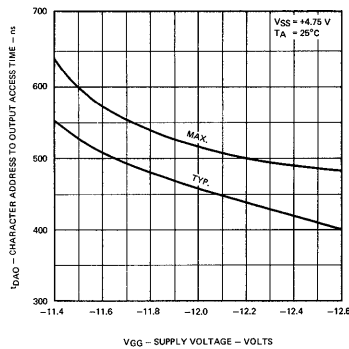


AC CHARACTERISTICS: $V_{SS} = +5 \text{ V} \pm 0.25 \text{ V}$, $V_{GG} = -12 \text{ V} \pm 0.6 \text{ V}$, $V_{DD} = 0 \text{ V}$, $C_L = 10 \text{ pF}$, $T_A = 0^\circ \text{C}$ to $+70^\circ \text{C}$.

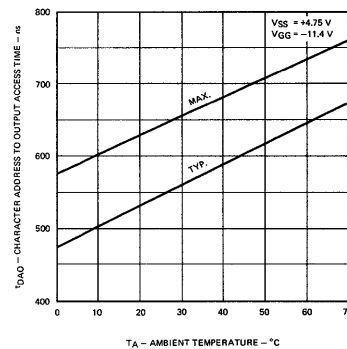
SYMBOL	CHARACTERISTIC	MIN.	TYP.	MAX.	UNITS	CONDITIONS
f	Clock Frequency	dc		1.0	MHz	
tPW ϕ	Clock Pulse Width	500			ns	
t _r , t _f	Clock Rise & Fall Time (10%-90%)			2.0	μs	
tPWR	Reset Pulse Width	500			ns	
tRS	Reset to Clock Set Up Time	100	300		ns	Fig. 5
tDAO	Character Address to Output Access Time		500	1000	ns	Notes 4 & 5, Fig. 1
tD ϕ O	Clock to Output Access Time		500	1000	ns	Notes 4 & 5, Fig. 2
tDRO	Reset to Output Time Delay		300	600	ns	Notes 4 & 5, Fig. 3
tDBO	Blanking to Output Time Delay		300	1000	ns	Notes 4 & 5, Fig. 6
tD ϕ C	Clock to Counter Output Time Delay		300	500	ns	Notes 4 & 5, Fig. 2
tDRC	Reset to Counter Output Time Delay		300	500	ns	Notes 4 & 5, Fig. 3
tDOE	Output Enable Delay Time		300	600	ns	Notes 4 & 5, Fig. 4
tDOD	Output Disable Delay Time		300	600	ns	Notes 4 & 5, Fig. 4
C _I	Input Capacitance		5.0	1.0	pF	f = 1.0 MHz, 0 V Bias Note 1

4. AC Output LOW level is defined as 0.4 V @ 1.6 mA, current sinking (i.e., 1 TTL load).
5. AC Output HIGH level is defined as 2.4 V @ -0.6 mA, current sourcing (i.e., 1 TTL load).

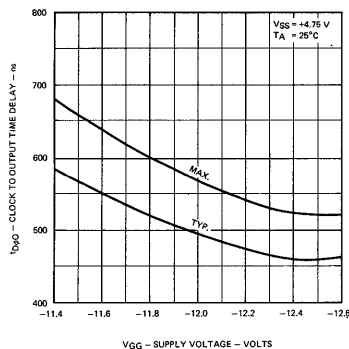
**CHARACTER ADDRESS TO OUTPUT
ACCESS TIME VERSUS
SUPPLY VOLTAGE**



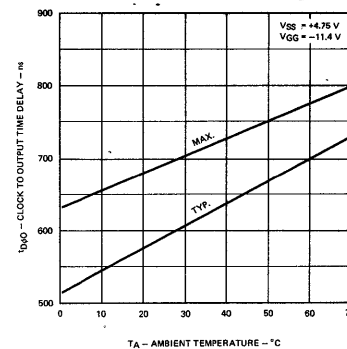
**CHARACTER ADDRESS TO OUTPUT
ACCESS TIME VERSUS
AMBIENT TEMPERATURE**



**CLOCK TO OUTPUT TIME DELAY
VERSUS
SUPPLY VOLTAGE**



**CLOCK TO OUTPUT TIME DELAY
VERSUS
AMBIENT TEMPERATURE**



TIMING DIAGRAM

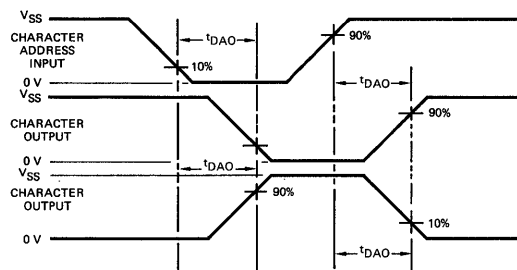


Fig. 1

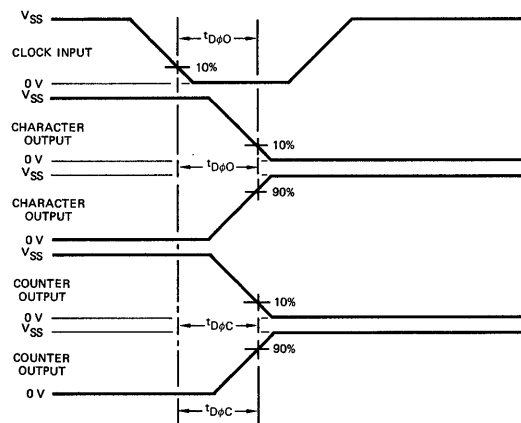


Fig. 2

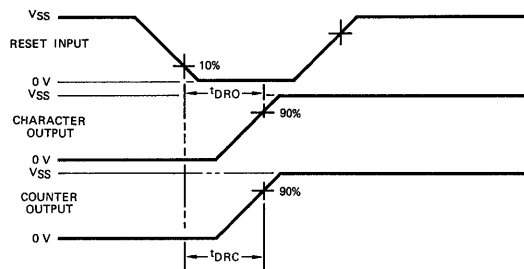


Fig. 3

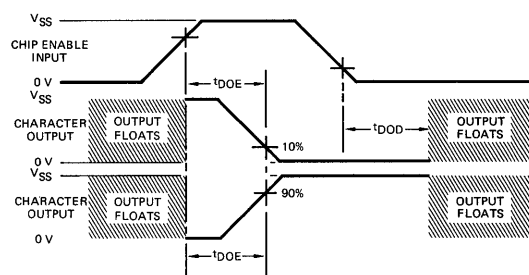


Fig. 4

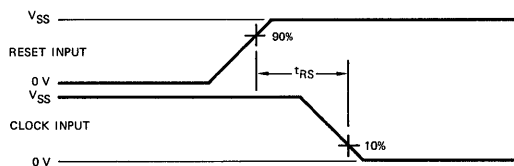


Fig. 5

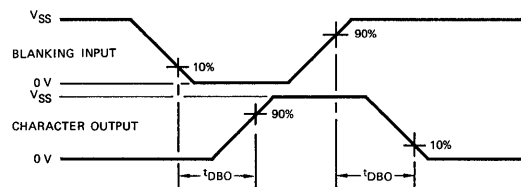
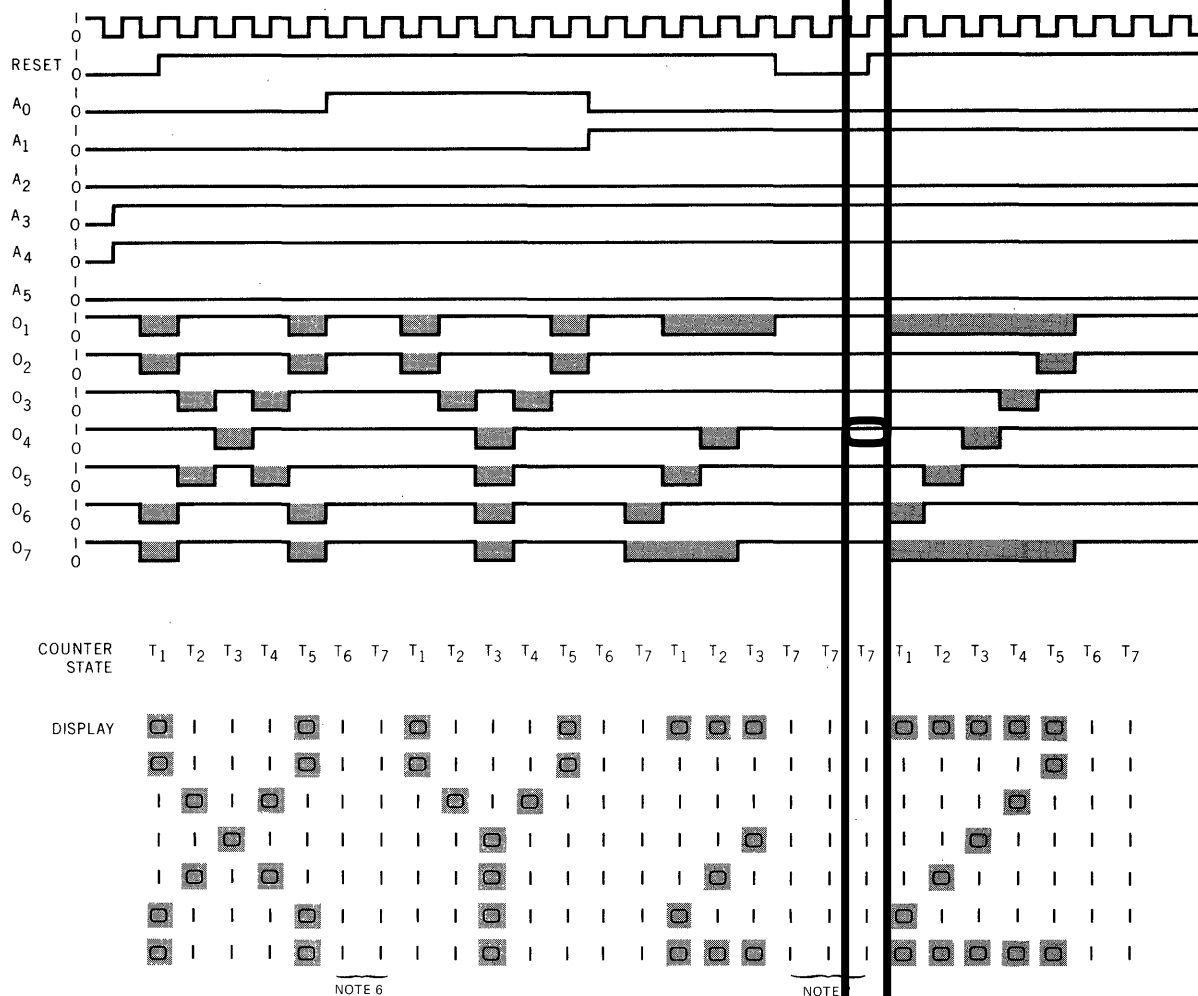


Fig. 6

GLOSSARY OF TERMS

1. V_{SS} The most positive voltage applied to the device.
2. V_{GG} The most negative voltage applied to the device
3. V_{DD} The next most negative voltage applied to the device.
4. t_{RS} Reset to clock set up time — reset must return to a logical HIGH within the period t_{RS}, before a positive to negative clock transition to insure character output levels during that clock time.
5. t_{DAO} Time for a character output to change logic levels after a character input changes logic levels.
6. t_{DφO} Time for a character output to change logic levels after the clock input reaches a logic LOW level.
7. t_{DRO} Time for a character output to reach a logic HIGH after reset reaches a logic LOW.
8. t_{DBO} Time for a character output to change logic levels after the blanking input changes logic levels.
9. t_{DφC} Time for the counter output to change logic levels after the clock input reaches a logic LOW level.
10. t_{DRC} Time for the counter output to reach a logic HIGH level after the reset input reaches a logic LOW level.
11. t_{DOE} Time before information is available at the character outputs after the chip enable input reaches a logic HIGH level.
12. t_{DOD} Time before the character outputs are disabled after the chip enable input reaches a logic LOW level.

TYPICAL FUNCTIONAL TIMING DIAGRAM



NOTES:

6. Last two counter states (count mode control = HIGH $\Rightarrow$ MOD 7) provide blanking.
7. Counter is Reset to the last state.

Fig. 7

INTERFACING

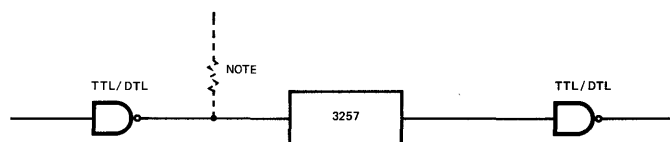
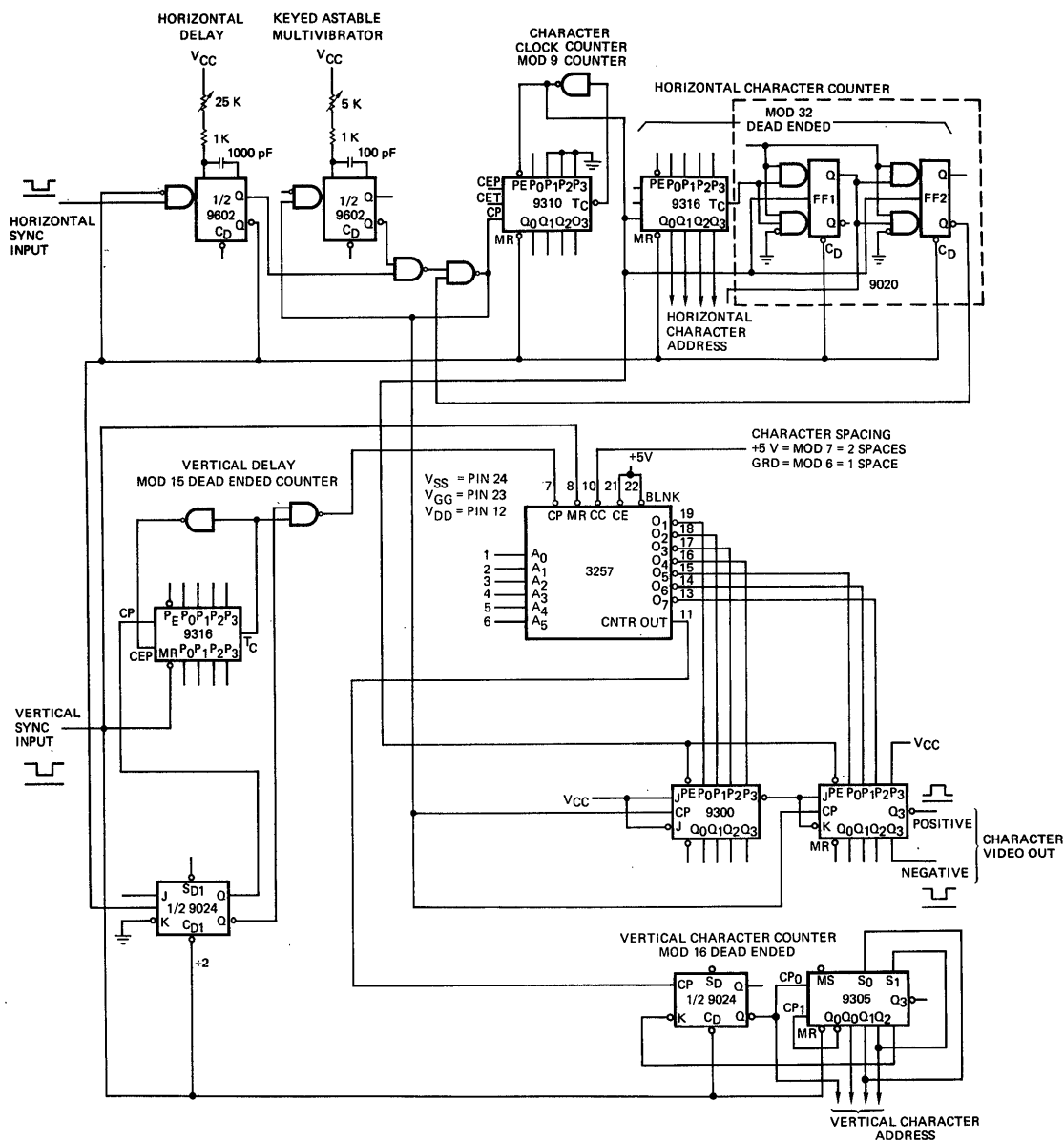


Fig. 8

Note: Directly compatible at outputs with TTL/DTL. Inputs directly compatible with DTL. When being driven by TTL, no pullup resistor needed if TTL output swings to ($V_{SS} - 1$) volts.



NOTE:
Horizontal and vertical are referred to as in a standard TV type raster.
16 characters per line, 32 character lines in system.
Each character 10 raster lines wide.

Fig. 9 TYPICAL VERTICAL RASTER SCAN APPLICATION

CUSTOM FONT

ORDERING INFORMATION

Additional patterns may be made available upon request. The 3257 is programmed on IBM cards or IBM forms in the coding format shown below:

A logic "1" = A more positive voltage nominally +5 V
A logic "0" = A more negative voltage nominally 0 V
The character "dots" are defined as logic "0"

6, 7, 8, 9, 10, 11
22, 23, 24, 25, 26, 27, 28
30, 31, 32, 33, 34, 35, 36
38, 39, 40, 41, 42, 43, 44
46, 47, 48, 49, 50, 51, 52
54, 55, 56, 57, 58, 59, 60
73, 74, 75, 76, 77, 78, 79, 80

Character address input code. The most significant bit (A5) is in column 11.
The first column of the character addressed. The most significant bit (07) is in Column 28.
The next column of the character addressed. The most significant bit (07) is in Column 36.
The next column of the character addressed. The most significant bit (07) is in Column 44.
The next column of the character addressed. The most significant bit (07) is in Column 52.
The last column of the character addressed. The most significant bit (07) is in Column 60.
Coding these columns is not essential and may be used for card identification purpose.

3257A – STANDARD ASCII CHARACTER FONT

COUNT MODE CONTROL $\cong$ GND $\Rightarrow$ MOD 6

A ₀	0	1	0	1	0	1	0	1
A ₁	0	0	1	1	0	0	1	1
A ₂	0	0	0	0	1	1	1	1

A ₅	A ₄	A ₃	O ₁	O ₂	O ₃	O ₄	O ₅	O ₆	O ₇	T ₁	T ₂	T ₃	T ₄	T ₅	T ₆
0	0	0													
0	0	1													
0	1	0													
0	1	1													
1	0	0													
1	0	1													
1	1	0													
1	1	1													

NOTE: Order 3257ADC for character font shown.